

Lecture 04 - Geometric Projection

This lecture explains how a 3D view becomes a 2D image. Projection is where camera geometry, homogeneous coordinates, clipping planes, depth precision, and viewport mapping meet.

Source chunk	course_text_parts/03_lectures/04_geometric-projection.txt
Extracted slide pages	61
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Untitled slide
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Professor-style explanation	For 'Untitled slide', I would not just read the bullet points aloud. I would ask what problem the slide is solving and how it connects to the previous and next stage. The slide gives us this anchor: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image. Turn the slide into a causal explanation: this input is processed by this idea, which produces this result, and that result matters later.
Technical commentary	This slide is about Untitled slide. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Untitled slide' into a causal sentence instead of repeating the slide title?
Page 2	Projecting 3D Models
Source cue	- To compute the 2D representation of a 3D model on the screen, / it needs to be projected, whereby it loses its z-coordinates / - Depending on the projection type, the object might get skewed / (perspective shortening) / - Projection can also be expressed as a matrix multiplication,

Professor-style explanation	For 'Projecting 3D Models', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - To compute the 2D representation of a 3D model on the screen, / it needs to be projected, whereby it loses its z-coordinates / - Depending on the projection type, the object might get skewed / (perspective shortening) / - Projection can also be expressed as a matrix multiplication,. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Projecting 3D Models. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - To compute the 2D representation of a 3D model on the screen, / it needs to be projected, whereby it loses its z-coordinates / - Depending on the projection type, the object might get skewed / (perspective shortening) / - Projection can also be expressed as a matrix multiplication,
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Projecting 3D Models'?
Page 3	4.1 Linear Perspective
Source cue	4.2 Planar Projections / 4.3 Camera Modeling / 4.4 Specifying Projections in OpenGL / 4.5 Perspective Projection Derivation / 4.6 Orthographic Projection Derivation
Professor-style explanation	On this slide, I would emphasize that OpenGL is a controlled state machine around the GPU pipeline. '4.1 Linear Perspective' is not about one magic render call; it is about objects, bindings, shader interfaces, buffers, and state being consistent at draw time. The slide gives us this anchor: 4.2 Planar Projections / 4.3 Camera Modeling / 4.4 Specifying Projections in OpenGL / 4.5 Perspective Projection Derivation / 4.6 Orthographic Projection Derivation. If the output is wrong, you inspect which object owns the data, which shader consumes it, and which state changes the result.
Technical commentary	This slide is about 4.1 Linear Perspective. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: 4.2 Planar Projections / 4.3 Camera Modeling / 4.4 Specifying Projections in OpenGL / 4.5 Perspective Projection Derivation / 4.6 Orthographic Projection Derivation
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '4.1 Linear Perspective'?
Page 4	4.1 Linear Perspective
Source cue	Discovery of perspective distortion in art

Professor-style explanation	This slide should be read as camera geometry. With '4.1 Linear Perspective', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: Discovery of perspective distortion in art. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about 4.1 Linear Perspective. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: Discovery of perspective distortion in art
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how '4.1 Linear Perspective' changes positions before rasterization?

Page 5	Linear Perspective
Source cue	- What is wrong in this image? / William Hogarth: / False Perspective, 1754 / Satire on False Perspective / William Hogarth (1753)
Professor-style explanation	This slide should be read as camera geometry. With 'Linear Perspective', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - What is wrong in this image? / William Hogarth: / False Perspective, 1754 / Satire on False Perspective / William Hogarth (1753). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Linear Perspective. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - What is wrong in this image? / William Hogarth: / False Perspective, 1754 / Satire on False Perspective / William Hogarth (1753)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Linear Perspective' changes positions before rasterization?

Page 6	Drawing by Projecting
Source cue	- Projection: method for imaging objects using parallel or central beams / - Perspective: / "The science which teaches how to represent tri-dimensional objects / on a bi-dimensional surface, so that the perspective image coincides with / the one which is given by direct vision."

Professor-style explanation	This slide should be read as camera geometry. With 'Drawing by Projecting', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Projection: method for imaging objects using parallel or central beams / - Perspective: / "The science which teaches how to represent tri-dimensional objects / on a bi-dimensional surface, so that the perspective image coincides with / the one which is given by direct vision.". Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Drawing by Projecting. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Projection: method for imaging objects using parallel or central beams / - Perspective: / "The science which teaches how to represent tri-dimensional objects / on a bi-dimensional surface, so that the perspective image coincides with / the one which is given by direct vision."
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Drawing by Projecting' changes positions before rasterization?
Page 7	Projection in the Arts 1/5
Source cue	- Before the Renaissance, pictures often had a religious / and symbolic character / - Perspective was not understood: / accordingly, they seem 2-dimensional / Fresco from an Egyptian grave
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 1/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Before the Renaissance, pictures often had a religious / and symbolic character / - Perspective was not understood: / accordingly, they seem 2-dimensional / Fresco from an Egyptian grave. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 1/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Before the Renaissance, pictures often had a religious / and symbolic character / - Perspective was not understood: / accordingly, they seem 2-dimensional / Fresco from an Egyptian grave
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 1/5' changes positions before rasterization?
Page 8	Projection in the Arts 2/5

Source cue	- Techniques to simulate space: / - Shaded round volumetric shapes / - Spatial depth through converging lines / Giotto di Bondone / (1266-1337, Florence)
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 2/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Techniques to simulate space: / - Shaded round volumetric shapes / - Spatial depth through converging lines / Giotto di Bondone / (1266-1337, Florence). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 2/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Techniques to simulate space: / - Shaded round volumetric shapes / - Spatial depth through converging lines / Giotto di Bondone / (1266-1337, Florence)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 2/5' changes positions before rasterization?

Page 9	Projection in the Arts 3/5
Source cue	- Renaissance (1400-1700) / - New emphasis on the importance of personal views and interpretation of / the world, power of observation, especially of nature / (astronomy, anatomy, botany, ...) / - Universe as a clockwork:
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 3/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Renaissance (1400-1700) / - New emphasis on the importance of personal views and interpretation of / the world, power of observation, especially of nature / (astronomy, anatomy, botany, ...) / - Universe as a clockwork. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 3/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Renaissance (1400-1700) / - New emphasis on the importance of personal views and interpretation of / the world, power of observation, especially of nature / (astronomy, anatomy, botany, ...) / - Universe as a clockwork:
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 3/5' changes positions before rasterization?

Page 10	Projection in the Arts 4/5
Source cue	- Examples from the Renaissance
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 4/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Examples from the Renaissance. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 4/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Examples from the Renaissance
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 4/5' changes positions before rasterization?
Page 11	Projection in the Arts 4/5
Source cue	- Examples from the Renaissance / The Last Supper / Leonardo da Vinci (1498)
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 4/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Examples from the Renaissance / The Last Supper / Leonardo da Vinci (1498). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 4/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Examples from the Renaissance / The Last Supper / Leonardo da Vinci (1498)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 4/5' changes positions before rasterization?
Page 12	Projection in the Arts 5/5
Source cue	- "The Invention of the Painting" (1830) by Karl Friedrich Schinkel (1781-1841): / - An image based on a legend by Pliny the Elder / - Tracing the shadow contours on a drawing surface

Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 5/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - "The Invention of the Painting" (1830) by Karl Friedrich Schinkel (1781-1841): / - An image based on a legend by Pliny the Elder / - Tracing the shadow contours on a drawing surface. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 5/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - "The Invention of the Painting" (1830) by Karl Friedrich Schinkel (1781-1841): / - An image based on a legend by Pliny the Elder / - Tracing the shadow contours on a drawing surface
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 5/5' changes positions before rasterization?

Page 13	Projection in the Arts 6/5
Source cue	- M. C. Escher (1898 – 1972)
Professor-style explanation	This slide should be read as camera geometry. With 'Projection in the Arts 6/5', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - M. C. Escher (1898 – 1972). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection in the Arts 6/5. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - M. C. Escher (1898 – 1972)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection in the Arts 6/5' changes positions before rasterization?

Page 14	Projection Technics 1/3
Source cue	- Filippo Brunelleschi (1377-1446): / Systematic method for obtaining / a perspective projection
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Technics 1/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Filippo Brunelleschi (1377-1446): / Systematic method for obtaining / a perspective projection. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.

Technical commentary	This slide is about Projection Technics 1/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Filippo Brunelleschi (1377-1446): / Systematic method for obtaining / a perspective projection
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Technics 1/3' changes positions before rasterization?

Page 15	Projection Technics 1/3
Source cue	The Delivery of the Keys / Perugino (1482)
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Technics 1/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: The Delivery of the Keys / Perugino (1482). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Technics 1/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: The Delivery of the Keys / Perugino (1482)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Technics 1/3' changes positions before rasterization?

Page 16	Projection Technics 2/3
Source cue	- Formal description of the perspective projection / - Leone Batista Alberti (1404-1472): / First essay on the subject of "La Pittura" (1435) / - "A painting [the projection plane] is the intersection of a visual pyramid [view / frustum] at a given distance, with a fixed center [center of projection] and a
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Technics 2/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Formal description of the perspective projection / - Leone Batista Alberti (1404-1472): / First essay on the subject of "La Pittura" (1435) / - "A painting [the projection plane] is the intersection of a visual pyramid [view / frustum] at a given distance, with a fixed center [center of projection] and a. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.

Technical commentary	This slide is about Projection Technics 2/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Formal description of the perspective projection / - Leone Batista Alberti (1404-1472): / First essay on the subject of "La Pittura" (1435) / - "A painting [the projection plane] is the intersection of a visual pyramid [view / frustum] at a given distance, with a fixed center [center of projection] and a
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Technics 2/3' changes positions before rasterization?

Page 17	Projection Technics 3/3
Source cue	- Albrecht Dürer (1471-1528): / Geometric and mechanical / description of the concept / of similar triangles / - Construction of several
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Technics 3/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Albrecht Dürer (1471-1528): / Geometric and mechanical / description of the concept / of similar triangles / - Construction of several. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Technics 3/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Albrecht Dürer (1471-1528): / Geometric and mechanical / description of the concept / of similar triangles / - Construction of several
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Technics 3/3' changes positions before rasterization?

Page 18	Projection Technics 3/3
Source cue	- Camera Obscura
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Technics 3/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Camera Obscura. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Technics 3/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Camera Obscura
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.

Check yourself	Can you explain how 'Projection Technics 3/3' changes positions before rasterization?
Page 19	4.2 Planar Projections
Source cue	Perspective and parallel projections
Professor-style explanation	This slide should be read as camera geometry. With '4.2 Planar Projections', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: Perspective and parallel projections. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about 4.2 Planar Projections. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: Perspective and parallel projections
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how '4.2 Planar Projections' changes positions before rasterization?
Page 20	Geometric Projections
Source cue	- Method of mapping objects onto a screen using projectors / - Planar geometric projections / - Projection beams are straight lines / - Projection surface is a plane / (image plane, picture plane,
Professor-style explanation	This slide should be read as camera geometry. With 'Geometric Projections', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Method of mapping objects onto a screen using projectors / - Planar geometric projections / - Projection beams are straight lines / - Projection surface is a plane / (image plane, picture plane,. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Geometric Projections. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Method of mapping objects onto a screen using projectors / - Planar geometric projections / - Projection beams are straight lines / - Projection surface is a plane / (image plane, picture plane,
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Geometric Projections' changes positions before rasterization?
Page 21	Projection Types 1/2
Source cue	- Perspective projection (= central projection) / - Established by projection center / (COP, center of projection) / - Perspective reduction: The size of the projection / of an object decreases with increasing distance

Professor-style explanation	This slide should be read as camera geometry. With 'Projection Types 1/2', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Perspective projection (= central projection) / - Established by projection center / (COP, center of projection) / - Perspective reduction: The size of the projection / of an object decreases with increasing distance. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Types 1/2. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Perspective projection (= central projection) / - Established by projection center / (COP, center of projection) / - Perspective reduction: The size of the projection / of an object decreases with increasing distance
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Types 1/2' changes positions before rasterization?

Page 22	Projection Types 2/2
Source cue	Planar geometric Projections / Parallel Perspective / Orthographic Oblique 1-Point / 2-Point / Top Axonometric
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Types 2/2', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: Planar geometric Projections / Parallel Perspective / Orthographic Oblique 1-Point / 2-Point / Top Axonometric. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Types 2/2. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: Planar geometric Projections / Parallel Perspective / Orthographic Oblique 1-Point / 2-Point / Top Axonometric
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Types 2/2' changes positions before rasterization?

Page 23	Orthographic Projections
Source cue	- Orthographic projections often used for / construction drawings of machines and parts / - Main views: Front view, top view and side view / - Advantages / - Accurate length and angle measurements possible

Professor-style explanation	This slide should be read as camera geometry. With 'Orthographic Projections', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Orthographic projections often used for / construction drawings of machines and parts / - Main views: Front view, top view and side view / - Advantages / - Accurate length and angle measurements possible. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Orthographic Projections. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Orthographic projections often used for / construction drawings of machines and parts / - Main views: Front view, top view and side view / - Advantages / - Accurate length and angle measurements possible
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Orthographic Projections' changes positions before rasterization?

Page 24	Axonometric Projections
Source cue	- Projection beams perpendicular to the projection plane / Isometric / - Projection plane not perpendicular to a coordinate axis / - Visible effects / - Several perpendicular surfaces visible
Professor-style explanation	For 'Axonometric Projections', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Projection beams perpendicular to the projection plane / Isometric / - Projection plane not perpendicular to a coordinate axis / - Visible effects / - Several perpendicular surfaces visible. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Axonometric Projections. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Projection beams perpendicular to the projection plane / Isometric / - Projection plane not perpendicular to a coordinate axis / - Visible effects / - Several perpendicular surfaces visible
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Axonometric Projections'?

Page 25	Isometric Projections
Source cue	- Applications in illustrations, furniture design, ... / - Advantages / - Communicate spatial impression in a single view / - Same length ratios along the main axes / - Disadvantages

Professor-style explanation	This slide should be read as camera geometry. With 'Isometric Projections', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Applications in illustrations, furniture design, ... / - Advantages / - Communicate spatial impression in a single view / - Same length ratios along the main axes / - Disadvantages. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Isometric Projections. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Applications in illustrations, furniture design, ... / - Advantages / - Communicate spatial impression in a single view / - Same length ratios along the main axes / - Disadvantages
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Isometric Projections' changes positions before rasterization?

Page 26	Oblique Projection 1/2
Source cue	- Projection rays not perpendicular to the projection plane / - Common: $\alpha = 30^\circ$ / - Cavalier projection $\alpha = 45^\circ$, i.e., $\tan(\alpha) = 1$ / a' projectors
Professor-style explanation	This slide should be read as camera geometry. With 'Oblique Projection 1/2', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Projection rays not perpendicular to the projection plane / - Common: $\alpha = 30^\circ$ / - Cavalier projection $\alpha = 45^\circ$, i.e., $\tan(\alpha) = 1$ / a' projectors. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Oblique Projection 1/2. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Projection rays not perpendicular to the projection plane / - Common: $\alpha = 30^\circ$ / - Cavalier projection $\alpha = 45^\circ$, i.e., $\tan(\alpha) = 1$ / a' projectors
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Oblique Projection 1/2' changes positions before rasterization?

Page 27	Oblique Projection 2/2
Source cue	- Advantages / - Can accurately represent the shape of a boundary surface / of an object for accurate measurements / - Missing foreshortening simplifies comparison of sizes / - Gives a good impression of the 3D appearance

Professor-style explanation	This slide should be read as camera geometry. With 'Oblique Projection 2/2', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Advantages / - Can accurately represent the shape of a boundary surface / of an object for accurate measurements / - Missing foreshortening simplifies comparison of sizes / - Gives a good impression of the 3D appearance. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Oblique Projection 2/2. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Advantages / - Can accurately represent the shape of a boundary surface / of an object for accurate measurements / - Missing foreshortening simplifies comparison of sizes / - Gives a good impression of the 3D appearance
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Oblique Projection 2/2' changes positions before rasterization?

Page 28	Properties of Parallel Projections
Source cue	- Assumption: Object surface of greatest interest lies parallel to a main plane / - Direction of projection: DOP (direction of projection) / - Normal of the projection plane: VPN (view plane normal) / - Top view (orthographic) / - VPN Main axis, DOP VPN
Professor-style explanation	This slide should be read as camera geometry. With 'Properties of Parallel Projections', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Assumption: Object surface of greatest interest lies parallel to a main plane / - Direction of projection: DOP (direction of projection) / - Normal of the projection plane: VPN (view plane normal) / - Top view (orthographic) / - VPN Main axis, DOP VPN. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Properties of Parallel Projections. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Assumption: Object surface of greatest interest lies parallel to a main plane / - Direction of projection: DOP (direction of projection) / - Normal of the projection plane: VPN (view plane normal) / - Top view (orthographic) / - VPN Main axis, DOP VPN
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Properties of Parallel Projections' changes positions before rasterization?

Page 29	Perspective Projections 1/3
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Source cue	- Properties / - Communicate realistic views of three-dimensional objects / and give a spatial impression / - Distort objects in perspective view / - Scaling of objects is lost
Professor-style explanation	This slide should be read as camera geometry. With 'Perspective Projections 1/3', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Properties / - Communicate realistic views of three-dimensional objects / and give a spatial impression / - Distort objects in perspective view / - Scaling of objects is lost. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Perspective Projections 1/3. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Properties / - Communicate realistic views of three-dimensional objects / and give a spatial impression / - Distort objects in perspective view / - Scaling of objects is lost
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Perspective Projections 1/3' changes positions before rasterization?
Page 30	Perspective Projections 2/3
Source cue	- Vanishing points / - For rectangular objects (e.g., axis parallel cubes) whose surface normals / are parallel to the x-, y-, or z-axis, the number of vanishing points is equal / to the number of coordinate axes intersected by the projection plane / 1-Point Perspective 2-Point Perspective 3-Point Perspective
Professor-style explanation	For 'Perspective Projections 2/3', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Vanishing points / - For rectangular objects (e.g., axis parallel cubes) whose surface normals / are parallel to the x-, y-, or z-axis, the number of vanishing points is equal / to the number of coordinate axes intersected by the projection plane / 1-Point Perspective 2-Point Perspective 3-Point Perspective. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Perspective Projections 2/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Vanishing points / - For rectangular objects (e.g., axis parallel cubes) whose surface normals / are parallel to the x-, y-, or z-axis, the number of vanishing points is equal / to the number of coordinate axes intersected by the projection plane / 1-Point Perspective 2-Point Perspective 3-Point Perspective
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.

Check yourself	Can you state the coordinate space before and after 'Perspective Projections 2/3'?
Page 31	Perspective Projections 3/3
Source cue	- What happens if the cube is rotated so that its surface normals are not / parallel to the coordinate axes? / - Although the projection plane intersects only one coordinate axis, in this case / you get 3 vanishing points
Professor-style explanation	For 'Perspective Projections 3/3', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - What happens if the cube is rotated so that its surface normals are not / parallel to the coordinate axes? / - Although the projection plane intersects only one coordinate axis, in this case / you get 3 vanishing points. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Perspective Projections 3/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - What happens if the cube is rotated so that its surface normals are not / parallel to the coordinate axes? / - Although the projection plane intersects only one coordinate axis, in this case / you get 3 vanishing points
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Perspective Projections 3/3'?
Page 32	4.3 Camera Modeling
Source cue	Orientation of the camera and definition of the view volume
Professor-style explanation	This slide should be read as camera geometry. With '4.3 Camera Modeling', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: Orientation of the camera and definition of the view volume. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about 4.3 Camera Modeling. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: Orientation of the camera and definition of the view volume
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how '4.3 Camera Modeling' changes positions before rasterization?
Page 33	Camera Modeling

Source cue	- Application of planar geometric projections, represented by 4×4 matrices, / to project a 3D scene onto a projection screen, i.e., the canvas / - Basic camera parameters / - Camera position / - Camera orientation
Professor-style explanation	This slide should be read as camera geometry. With 'Camera Modeling', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Application of planar geometric projections, represented by 4×4 matrices, / to project a 3D scene onto a projection screen, i.e., the canvas / - Basic camera parameters / - Camera position / - Camera orientation. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Camera Modeling. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Application of planar geometric projections, represented by 4×4 matrices, / to project a 3D scene onto a projection screen, i.e., the canvas / - Basic camera parameters / - Camera position / - Camera orientation
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Camera Modeling' changes positions before rasterization?

Page 34	View Volume
Source cue	- Objects that are within the view volume are visible for the camera / - Conical viewing volume / - Clipping the scene objects against a conical viewing volume is expensive / (solving quadratic equations) / - Frustum - rectangular viewing volume
Professor-style explanation	This slide should be read as camera geometry. With 'View Volume', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Objects that are within the view volume are visible for the camera / - Conical viewing volume / - Clipping the scene objects against a conical viewing volume is expensive / (solving quadratic equations) / - Frustum - rectangular viewing volume. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about View Volume. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Objects that are within the view volume are visible for the camera / - Conical viewing volume / - Clipping the scene objects against a conical viewing volume is expensive / (solving quadratic equations) / - Frustum - rectangular viewing volume
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'View Volume' changes positions before rasterization?

Page 35	Projection Specification
Source cue	- Specified by Height Angle, Aspect Ratio, / Near Plane Distance and Far Plane Distance / - Examples of Aspect Ratio / - NTSC 4 : 3, HDTV 16 : 9 / - Cinema 2.35 : 1, 2.39 : 1, 2.40 : 1, or 2.55 : 1
Professor-style explanation	This slide should be read as camera geometry. With 'Projection Specification', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Specified by Height Angle, Aspect Ratio, / Near Plane Distance and Far Plane Distance / - Examples of Aspect Ratio / - NTSC 4 : 3, HDTV 16 : 9 / - Cinema 2.35 : 1, 2.39 : 1, 2.40 : 1, or 2.55 : 1. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Projection Specification. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Specified by Height Angle, Aspect Ratio, / Near Plane Distance and Far Plane Distance / - Examples of Aspect Ratio / - NTSC 4 : 3, HDTV 16 : 9 / - Cinema 2.35 : 1, 2.39 : 1, 2.40 : 1, or 2.55 : 1
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Projection Specification' changes positions before rasterization?

Page 36	View Angle
Source cue	- Parallel projection: no viewing angle as rays are parallel / - Perspective projection: / viewing angle determines perspective distortion in the image / - Specification / - Through the viewing angle in height (field of view in y, fovy)
Professor-style explanation	This slide should be read as camera geometry. With 'View Angle', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Parallel projection: no viewing angle as rays are parallel / - Perspective projection: / viewing angle determines perspective distortion in the image / - Specification / - Through the viewing angle in height (field of view in y, fovy). Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about View Angle. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Parallel projection: no viewing angle as rays are parallel / - Perspective projection: / viewing angle determines perspective distortion in the image / - Specification / - Through the viewing angle in height (field of view in y, fovy)
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'View Angle' changes positions before rasterization?

Page 37	Clipping Planes
Source cue	- Position of the planes is determined by the distance along the look vector: / near distance and far distance / - Near plane hides objects that are behind or too close to the camera / - Far plane hides objects that are too far away (and may be hard to spot) / $\Rightarrow$ Reduction of rendering overhead
Professor-style explanation	This slide should be read as camera geometry. With 'Clipping Planes', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Position of the planes is determined by the distance along the look vector: / near distance and far distance / - Near plane hides objects that are behind or too close to the camera / - Far plane hides objects that are too far away (and may be hard to spot) / $\Rightarrow$ Reduction of rendering overhead. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Clipping Planes. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Position of the planes is determined by the distance along the look vector: / near distance and far distance / - Near plane hides objects that are behind or too close to the camera / - Far plane hides objects that are too far away (and may be hard to spot) / $\Rightarrow$ Reduction of rendering overhead
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Clipping Planes' changes positions before rasterization?
Page 38	4.4 Specifying Projections in OpenGL
Source cue	Functions to describe the view volume
Professor-style explanation	On this slide, I would emphasize that OpenGL is a controlled state machine around the GPU pipeline. '4.4 Specifying Projections in OpenGL' is not about one magic render call; it is about objects, bindings, shader interfaces, buffers, and state being consistent at draw time. The slide gives us this anchor: Functions to describe the view volume. If the output is wrong, you inspect which object owns the data, which shader consumes it, and which state changes the result.
Technical commentary	This slide is about 4.4 Specifying Projections in OpenGL. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: Functions to describe the view volume
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in '4.4 Specifying Projections in OpenGL'?
Page 39	Projection Matrix Generation 1/3
Source cue	- Convenient generation of perspective projection matrix with GLM / - <code>glm::perspective(fovy, aspect, near, far) / h -n -f / aspect ■ / fovy fovy</code>

Professor-style explanation	For 'Projection Matrix Generation 1/3', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Convenient generation of perspective projection matrix with GLM / - glm::perspective(fovy, aspect, near, far) / h -n -f / aspect ■ / fovy fovy. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Projection Matrix Generation 1/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Convenient generation of perspective projection matrix with GLM / - glm::perspective(fovy, aspect, near, far) / h -n -f / aspect ■ / fovy fovy
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Projection Matrix Generation 1/3'?
Page 40	Projection Matrix Generation 2/3
Source cue	- Alternative generation of perspective projection matrix with GLM / - glm::frustum(left, right, bottom, top, near, far) / (r, t, -n) / z = -f (far plane) / (l, b, -n)
Professor-style explanation	For 'Projection Matrix Generation 2/3', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Alternative generation of perspective projection matrix with GLM / - glm::frustum(left, right, bottom, top, near, far) / (r, t, -n) / z = -f (far plane) / (l, b, -n). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Projection Matrix Generation 2/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Alternative generation of perspective projection matrix with GLM / - glm::frustum(left, right, bottom, top, near, far) / (r, t, -n) / z = -f (far plane) / (l, b, -n)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Projection Matrix Generation 2/3'?
Page 41	Projection Matrix Generation 3/3
Source cue	- Implementation of glm::perspective / detail::tmat4x4<T> perspective(T const & fovy, T const & aspect, / T const & near, T const & far) { / T const & l, r, b, t; / t = n * tan(fovy * M_PI / 360.0);

Professor-style explanation	For 'Projection Matrix Generation 3/3', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Implementation of glm::perspective / detail::tmat4x4<T> perspective(T const & fovy, T const & aspect, / T const & near, T const & far) { / T const & l, r, b, t; / t = n * tan(fovy * M_PI / 360.0). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Projection Matrix Generation 3/3. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Implementation of glm::perspective / detail::tmat4x4<T> perspective(T const & fovy, T const & aspect, / T const & near, T const & far) { / T const & l, r, b, t; / t = n * tan(fovy * M_PI / 360.0);
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Projection Matrix Generation 3/3'?
Page 42	Orthographic Matrix Specification
Source cue	- Convenient generation of parallel projection matrix with GLM / - glm::ortho(left, right, bottom, top, near, far) / (r, t, -f) / z = -f (far plane) / z = -n (near plane)
Professor-style explanation	For 'Orthographic Matrix Specification', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Convenient generation of parallel projection matrix with GLM / - glm::ortho(left, right, bottom, top, near, far) / (r, t, -f) / z = -f (far plane) / z = -n (near plane). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Orthographic Matrix Specification. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Convenient generation of parallel projection matrix with GLM / - glm::ortho(left, right, bottom, top, near, far) / (r, t, -f) / z = -f (far plane) / z = -n (near plane)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Orthographic Matrix Specification'?
Page 43	4.5 Orthographic Projection Derivation
Source cue	Step-wise transformation of a cuboid into a cube

Professor-style explanation	For '4.5 Orthographic Projection Derivation', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: Step-wise transformation of a cuboid into a cube. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about 4.5 Orthographic Projection Derivation. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Step-wise transformation of a cuboid into a cube
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '4.5 Orthographic Projection Derivation'?

Page 44	Canonical View Volume
Source cue	- Canonical view volume – simplest possible view volume / - Cube centered at the origin / - Cube faces $x = \pm 1$, $y = \pm 1$, $z = \pm 1$ / - Defined in a left-handed coordinate system (LHS) / (1, 1, 1)
Professor-style explanation	For 'Canonical View Volume', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Canonical view volume – simplest possible view volume / - Cube centered at the origin / - Cube faces $x = \pm 1$, $y = \pm 1$, $z = \pm 1$ / - Defined in a left-handed coordinate system (LHS) / (1, 1, 1). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Canonical View Volume. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Canonical view volume – simplest possible view volume / - Cube centered at the origin / - Cube faces $x = \pm 1$, $y = \pm 1$, $z = \pm 1$ / - Defined in a left-handed coordinate system (LHS) / (1, 1, 1)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Canonical View Volume'?

Page 45	Orthographic Projection 1/4
Source cue	- Transform right-handed view volume specified as $[-1, 1] \times [-1, 1] \times [-1, 1]$ / into left-handed canonical view volume $-1, 1 \times -1, 1 \times -1, 1$ / and then create the projected image by neglecting z-values / (r, t, -f) / (1, 1, 1)

Professor-style explanation	For 'Orthographic Projection 1/4', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Transform right-handed view volume specified as $[-1, 1] \times [-1, 1] \times [-1, 1]$ / into left-handed canonical view volume $[-1, 1] \times [-1, 1] \times [-1, 1]$ / and then create the projected image by neglecting z-values / $(r, t, -f) / (1, 1, 1)$. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Orthographic Projection 1/4. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Transform right-handed view volume specified as $[-1, 1] \times [-1, 1] \times [-1, 1]$ / into left-handed canonical view volume $[-1, 1] \times [-1, 1] \times [-1, 1]$ / and then create the projected image by neglecting z-values / $(r, t, -f) / (1, 1, 1)$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Orthographic Projection 1/4'?
Page 46	Orthographic Projection 2/4
Source cue	- Let (x, y, z) be a model point and (x', y', z') its transformed point / - The desired pixel (x, y) results from orthographic projection of (x', y', z') / onto the plane $z = -1$: $x = x'$ and $y = y'$ /
Professor-style explanation	For 'Orthographic Projection 2/4', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Let (x, y, z) be a model point and (x', y', z') its transformed point / - The desired pixel (x, y) results from orthographic projection of (x', y', z') / onto the plane $z = -1$: $x = x'$ and $y = y'$ / . The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Orthographic Projection 2/4. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Let (x, y, z) be a model point and (x', y', z') its transformed point / - The desired pixel (x, y) results from orthographic projection of (x', y', z') / onto the plane $z = -1$: $x = x'$ and $y = y'$ /
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Orthographic Projection 2/4'?
Page 47	Orthographic Projection 3/4
Source cue	- Step 2, scale cuboid from step 1 into cube / - Original cuboid bounds are $[-1, 1] \times [-1, 1] \times [-1, 1]$ / , $x = -1$, $x = 1$, $y = -1$, $y = 1$, $z = -1$, $z = 1$ /

Professor-style explanation	This slide should be read as camera geometry. With 'Orthographic Projection 3/4', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Step 2, scale cuboid from step 1 into cube / - Original cuboid bounds are $/ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} / -$, $\times -$, $\times -$, $/ 2 \ 2 \ 2 \ 2 \ 2$. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Orthographic Projection 3/4. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Step 2, scale cuboid from step 1 into cube / - Original cuboid bounds are $/ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} / -$, $\times -$, $\times -$, $/ 2 \ 2 \ 2 \ 2 \ 2$
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Orthographic Projection 3/4' changes positions before rasterization?

Page 48	Orthographic Projection 4/4
Source cue	- Multiplication of all orthographic projection matrices results in $/ 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} / 0 \ 0 - / \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} / 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$
Professor-style explanation	This slide should be read as camera geometry. With 'Orthographic Projection 4/4', the question is how a 3D view becomes coordinates that can be clipped, divided, mapped to the viewport, and rasterized. The slide gives us this anchor: - Multiplication of all orthographic projection matrices results in $/ 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} / 0 \ 0 - / \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} / 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$. Keep separate the camera/view transform, the projection matrix, the perspective divide, and the final viewport transform; many mistakes come from blending these steps together.
Technical commentary	This slide is about Orthographic Projection 4/4. Read it as camera geometry. Track how 3D view-space positions become clip coordinates, normalized device coordinates, and finally screen locations. The visible cue is: - Multiplication of all orthographic projection matrices results in $/ 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} / 0 \ 0 - / \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} / 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$
Why it matters	Projection controls both image composition and depth precision, so it affects visibility and rasterization later.
Check yourself	Can you explain how 'Orthographic Projection 4/4' changes positions before rasterization?

Page 49	4.6 Perspective Projection Derivation
Source cue	Step-wise transformation of a frustum into a cube
Professor-style explanation	For '4.6 Perspective Projection Derivation', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: Step-wise transformation of a frustum into a cube. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.

Technical commentary	This slide is about 4.6 Perspective Projection Derivation. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Step-wise transformation of a frustum into a cube
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '4.6 Perspective Projection Derivation'?
Page 50	Perspective Projection
Source cue	- Transform view volume created by <code>glm::frustum</code> or <code>glm::perspective</code> into / canonical view volume / and then create the projected image by neglecting z-values / (1, 1, 1) / (-1,-1,-1)
Professor-style explanation	For 'Perspective Projection', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Transform view volume created by <code>glm::frustum</code> or <code>glm::perspective</code> into / canonical view volume / and then create the projected image by neglecting z-values / (1, 1, 1) / (-1,-1,-1). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Perspective Projection. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Transform view volume created by <code>glm::frustum</code> or <code>glm::perspective</code> into / canonical view volume / and then create the projected image by neglecting z-values / (1, 1, 1) / (-1,-1,-1)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Perspective Projection'?
Page 51	Transformation of z-Values 1/2
Source cue	- After the perspective projection, z-values shall be mapped as follows / - All z-values lying on the near plane shall have a value of -1 / - All z-values lying on the far plane shall have a value of 1 / - The following matrix ■ allows modification of the depth values / and assures the flip (left-handed to right-handed conversion)

Professor-style explanation	For 'Transformation of z-Values 1/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - After the perspective projection, z-values shall be mapped as follows / - All z-values lying on the near plane shall have a value of -1 / - All z-values lying on the far plane shall have a value of 1 / - The following matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ allows modification of the depth values / and assures the flip (left-handed to right-handed conversion). The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Transformation of z-Values 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - After the perspective projection, z-values shall be mapped as follows / - All z-values lying on the near plane shall have a value of -1 / - All z-values lying on the far plane shall have a value of 1 / - The following matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ allows modification of the depth values / and assures the flip (left-handed to right-handed conversion)
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Transformation of z-Values 1/2'?

Page 52	Transformation of z-Values 2/2
Source cue	- Choosing values for a and b in $\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ must fulfill the following / $-\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ / $-\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Professor-style explanation	For 'Transformation of z-Values 2/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Choosing values for a and b in $\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ must fulfill the following / $-\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Transformation of z-Values 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Choosing values for a and b in $\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ must fulfill the following / $-\begin{bmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Transformation of z-Values 2/2'?

Page 53	Transformation of x- and y-Values 1/2
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Source cue	- Derivation so far leaves us with the following matrix / which only changes z-coordinates / $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ / - $\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z + 2 \\ 1 \end{bmatrix}$
Professor-style explanation	For 'Transformation of x- and y-Values 1/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Derivation so far leaves us with the following matrix / which only changes z-coordinates / $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ / - $\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z + 2 \\ 1 \end{bmatrix}$. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Transformation of x- and y-Values 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Derivation so far leaves us with the following matrix / which only changes z-coordinates / $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ / - $\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z + 2 \\ 1 \end{bmatrix}$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Transformation of x- and y-Values 1/2'?

Page 54	Transformation of x- and y-Values 2/2
Source cue	- To transform p to the coordinates (1,0,-1), the following equation must hold / $\frac{1}{\sqrt{2}} \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan^2 \theta / 2 \cdot 2 \cdot 1$
Professor-style explanation	For 'Transformation of x- and y-Values 2/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - To transform p to the coordinates (1,0,-1), the following equation must hold / $\frac{1}{\sqrt{2}} \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan^2 \theta / 2 \cdot 2 \cdot 1$. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Transformation of x- and y-Values 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - To transform p to the coordinates (1,0,-1), the following equation must hold / $\frac{1}{\sqrt{2}} \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan \theta / \frac{1}{\sqrt{2}} \cdot \tan^2 \theta / 2 \cdot 2 \cdot 1$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Transformation of x- and y-Values 2/2'?

Page 55	4.7 Viewport Transformation
Source cue	Mapping the normalized device coordinates to the screen

Professor-style explanation	For '4.7 Viewport Transformation', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: Mapping the normalized device coordinates to the screen. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about 4.7 Viewport Transformation. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: Mapping the normalized device coordinates to the screen
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after '4.7 Viewport Transformation'?

Page 56	Screen Coordinate Mapping
Source cue	- Mapping of the viewing plane on a drawing surface / - Define the pixel rectangle to be mapped to / - Initially, this rectangle is the size of the entire canvas / - <code>glViewport(GLint x, GLint y, GLsizei width, GLsizei height)</code> / - When window size changes, adjust the aspect ratio to avoid distortion
Professor-style explanation	For 'Screen Coordinate Mapping', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Mapping of the viewing plane on a drawing surface / - Define the pixel rectangle to be mapped to / - Initially, this rectangle is the size of the entire canvas / - <code>glViewport(GLint x, GLint y, GLsizei width, GLsizei height)</code> / - When window size changes, adjust the aspect ratio to avoid distortion. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Screen Coordinate Mapping. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Mapping of the viewing plane on a drawing surface / - Define the pixel rectangle to be mapped to / - Initially, this rectangle is the size of the entire canvas / - <code>glViewport(GLint x, GLint y, GLsizei width, GLsizei height)</code> / - When window size changes, adjust the aspect ratio to avoid distortion
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Screen Coordinate Mapping'?

Page 57	Viewport Transformation 1/2
Source cue	- Specification of the viewport parameters by <code>glm::perspective</code> / Projection / <code>glm::frustum</code> / transformation / - <code>glViewport(x, y, w, h)</code> <code>glm::ortho</code>

Professor-style explanation	For 'Viewport Transformation 1/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Specification of the viewport parameters by glm::perspective / Projection / glm::frustum / transformation / - glViewport(x, y, w, h) glm::ortho. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Viewport Transformation 1/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Specification of the viewport parameters by glm::perspective / Projection / glm::frustum / transformation / - glViewport(x, y, w, h) glm::ortho
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Viewport Transformation 1/2'?

Page 58	Viewport Transformation 2/2
Source cue	- Viewport transformation is a 2D transformation defined as $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ / $\Delta x \Delta y$ / - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ / $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
Professor-style explanation	For 'Viewport Transformation 2/2', put the formula aside for a moment and name the coordinate spaces. A transformation only makes sense when we know where the point, vector, or normal starts and where it should end. The slide gives us this anchor: - Viewport transformation is a 2D transformation defined as $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ / $\Delta x \Delta y$ / - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ / $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$. The professor-level way to read this slide is to narrate the movement: model space to world space, world to view, view to clip, or whichever step the slide is showing.
Technical commentary	This slide is about Viewport Transformation 2/2. Read it as a coordinate-space operation. Name the input space, the matrix or transformation, and the output space before memorizing formulas. The visible cue is: - Viewport transformation is a 2D transformation defined as $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ / $\Delta x \Delta y$ / - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ / $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
Why it matters	A wrong coordinate-space assumption can make correct formulas produce wrong images.
Check yourself	Can you state the coordinate space before and after 'Viewport Transformation 2/2'?

Page 59	volume, we can differentiate different projection types
Source cue	- Projection has its origins in the arts / - Based on the deformation of the initial view volume to the canonical view / volume, we can differentiate different projection types / - In OpenGL two main projection types exist / - Perspective projection results in perspective shortening

Professor-style explanation	On this slide, I would emphasize that OpenGL is a controlled state machine around the GPU pipeline. 'volume, we can differentiate different projection types' is not about one magic render call; it is about objects, bindings, shader interfaces, buffers, and state being consistent at draw time. The slide gives us this anchor: - Projection has its origins in the arts / - Based on the deformation of the initial view volume to the canonical view / volume, we can differentiate different projection types / - In OpenGL two main projection types exist / - Perspective projection results in perspective shortening. If the output is wrong, you inspect which object owns the data, which shader consumes it, and which state changes the result.
Technical commentary	This slide is about volume, we can differentiate different projection types. Read it as concrete API state and GPU data movement. Ask which object is bound, which shader stage consumes it, and which state affects the draw call. The visible cue is: - Projection has its origins in the arts / - Based on the deformation of the initial view volume to the canonical view / volume, we can differentiate different projection types / - In OpenGL two main projection types exist / - Perspective projection results in perspective shortening
Why it matters	OpenGL bugs are usually state, binding, shader-interface, or buffer-layout bugs, so API details matter.
Check yourself	Can you name the OpenGL object, state, shader stage, or buffer involved in 'volume, we can differentiate different projection types'?
Page 60	Literature and other sources used in this chapter
Source cue	The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Professor-style explanation	For 'Literature and other sources used in this chapter', I would not just read the bullet points aloud. I would ask what problem the slide is solving and how it connects to the previous and next stage. The slide gives us this anchor: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image. Turn the slide into a causal explanation: this input is processed by this idea, which produces this result, and that result matters later.
Technical commentary	This slide is about Literature and other sources used in this chapter. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: The extracted slide text is mostly visual or metadata; use the original PDF page for the diagram or image.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?
Page 61	- Text Books
Source cue	- J. Foley, A. van Dam, S. Feiner: Computer Graphics: Principles and Practice (3rd / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics

Professor-style explanation	For '- Text Books', I would not just read the bullet points aloud. I would ask what problem the slide is solving and how it connects to the previous and next stage. The slide gives us this anchor: - J. Foley, A. van Dam, S. Feiner: Computer Graphics: Principles and Practice (3rd / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics. Turn the slide into a causal explanation: this input is processed by this idea, which produces this result, and that result matters later.
Technical commentary	This slide is about - Text Books. Connect the bullet terms causally: what problem is being solved, what data is used, what output is produced, and which later pipeline stage depends on it. The visible cue is: - J. Foley, A. van Dam, S. Feiner: Computer Graphics: Principles and Practice (3rd / - P. Shirley, M. Ashikhmin, S. Marschner: Fundamentals of Computer Graphics
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn '- Text Books' into a causal sentence instead of repeating the slide title?

04.1 Linear Perspective and Planar Projections

Source location: Sections 4.1-4.2

Commentary

Perspective projection models the visual effect that farther objects appear smaller. Orthographic projection removes this distance-based size change and keeps parallel lines parallel. Both are useful, but they communicate different spatial relationships.

Planar projection means mapping points onto an image plane. The lecture's art examples are not decoration; they show that projection is a geometric rule for constructing an image from a viewpoint.

Mental model

Projection is the rule that turns 3D positions into image-plane positions.

Check yourself

What visual cue does perspective projection add that orthographic projection lacks?

04.2 Camera Modeling

Source location: Section 4.3

Commentary

A virtual camera is defined by position, orientation, projection type, viewing volume, and image/viewport settings. The view transform places the world relative to the camera. The projection transform maps that view into clip coordinates.

Near and far clipping planes are part of the camera model. They decide what range of depths is represented and strongly affect depth buffer precision. A careless near/far setup can create z-fighting even if the scene geometry is correct.

Mental model

A camera is not only an eye position; it is also a volume and a mapping rule.

Check yourself

Why can changing the near plane affect depth artifacts?

04.3 Specifying Projections in OpenGL

Source location: Section 4.4

Commentary

In OpenGL, projection is usually encoded in a projection matrix sent to a shader. The matrix maps view-space coordinates to clip space. Clipping and the perspective divide then lead toward normalized device coordinates.

Aspect ratio and field of view must match the intended viewport. If the aspect ratio is wrong, objects appear stretched. If the field of view is extreme, the image can look distorted even though the math is functioning.

Mental model

The projection matrix is the camera lens encoded as linear algebra in homogeneous coordinates.

Check yourself

What symptom suggests that the projection aspect ratio does not match the window?

04.4 Orthographic and Perspective Derivations

Source location: Sections 4.5-4.6

Commentary

The derivations are there to explain where the matrix entries come from. Orthographic projection maps a box-shaped view volume into normalized coordinates. Perspective projection maps a frustum and uses the homogeneous component for the later divide.

You do not need to treat these matrices as random formulas. Each part maps a coordinate range into a standard range. The perspective matrix additionally arranges w so that the perspective divide creates foreshortening.

Mental model

Projection matrices normalize a viewing volume; perspective also prepares the divide by w .

Check yourself

What does the perspective divide do to x and y coordinates?

04.5 Viewport Transformation

Source location: Section 4.7

Commentary

After normalized device coordinates, the viewport transform maps the normalized range to actual window coordinates. This is the last geometric mapping before rasterization uses the target grid.

Separating projection from viewport mapping helps debugging. A wrong projection can produce wrong normalized coordinates; a wrong viewport can map otherwise valid coordinates into the wrong part of the window.

Mental model

Projection decides normalized position; viewport decides where that position lands on the screen.

Check yourself

Why is resizing a window related to both viewport and projection settings?

End-of-lecture summary

If you remember only one thing from Lecture 04, remember this: This lecture explains how a 3D view becomes a 2D image. Projection is where camera geometry, homogeneous coordinates, clipping planes, depth precision, and viewport mapping meet.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.